
ELE302 Assignment Report: Predicting Household Energy Demand Using Regression and Ensemble Methods

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1 — Background and Motivation

As electricity demand increases in the modern world, power distribution companies offer a range of services to forecast energy consumption and manage demand. This includes providing and installing a household energy smart meter for the local householder to alleviate pressure on the local power grid. To successfully predict local energy consumption, engineers must not only ensure the smart meter's accuracy and reliability but also develop feasible, practical mathematical models. This report uses electrical-level and meteorological data as proxy indicators to predict grid energy demand using machine learning algorithms.

The machine learning method significantly improves the ability to handle large, complex datasets by automatically discovering complex patterns and reducing reliance on manual analysis and derivation. Especially when dealing with variables with unknown or non-linear relationships, machine learning not only reduces the workload for data scientists but also improves the accuracy of the predictors and their versatility for real-world deployments — qualities unmatched by traditional algorithms.

2 — Data Cleaning and Processing (Task 1)

2.1 — Project objectives

The given dataset has a total of 50390 (plus a title row) data rows and 30 variable columns containing 1 target (*EnergyRequestedFromGrid*) and 29 feature variables. Within the 30 variables, most of them are numerical data (excluding *WeatherIcon*). The objectives are to use MATLAB to clean and preprocess the data, build a linear regression (LR) model and a second model to compare their performance, and finally summarise the selected approaches and discuss ethical issues and future machine learning challenges.

2.2 Missing Data and Outliers Treatment

Prior to the feature selection process, missing values were identified first using `isnan` and `isinf` functions to ensure there are no incomplete variables propagated into the modelling pipeline. This step is essential because MATLAB can not process strings directly before converting them into numbers (*WeatherIcons* contain strings only, see section 2.3). After the calculation, there are a total of 21 NaN and 2 Inf in the raw dataset.

There are several methods to convert these undesired variables to numerical data, including deletion (removing the whole row), replaced by the mean and median. However, the deletion method was rejected because it reduces the sampling size, including removing other suitable data for training. In contrast with replacing by median, replacing by mean will cause outliers to create a false correlation if the dataset contains a few extreme values, because every value contributes equally to the result. Therefore, a more robust method, such as replacing with the median is more suitable because a few individual extreme values will not affect the overall results.

Similarly, outliers can also be replaced by the median. However, outlier detection is more complicated and challenging compared with replacing missing data. The selection approach uses the Z-score to detect outliers, it measures how many standard deviations a value is located

from the column mean as $\mathbf{Z} = (\mathbf{x} - \boldsymbol{\mu}) / \boldsymbol{\sigma}$. The selected approach uses an iterative method ($Z = 3 - 15$, interval of 1), combining charts, R^2 and RMSE outputs to validate the optimal Z value to improve the accuracy of the final decision making.

2.3 Features Selection

Prior to the missing data and outliers treatment, *WeatherIcon* was first removed because it doesn't contain physical measurements; only categorical weather classifications are insufficient to demonstrate the correlation between weather and power grid energy consumption [1]. Moreover, since it also contains string contents, which means one-hot encoding is no longer required. Feature selection is then performed to reduce dimensionality, remove noise, and prevent the model from overfitting. Pearson correlation coefficients were used to compute the correlation between each feature and the target variables before standardisation to avoid data leakage, resulting in a ranked bar chart shown in *Figure 1*. Features are removed primarily due to physical redundancy; the detailed reasons for rejecting each feature are documented in the code comments of *main_run.m*.

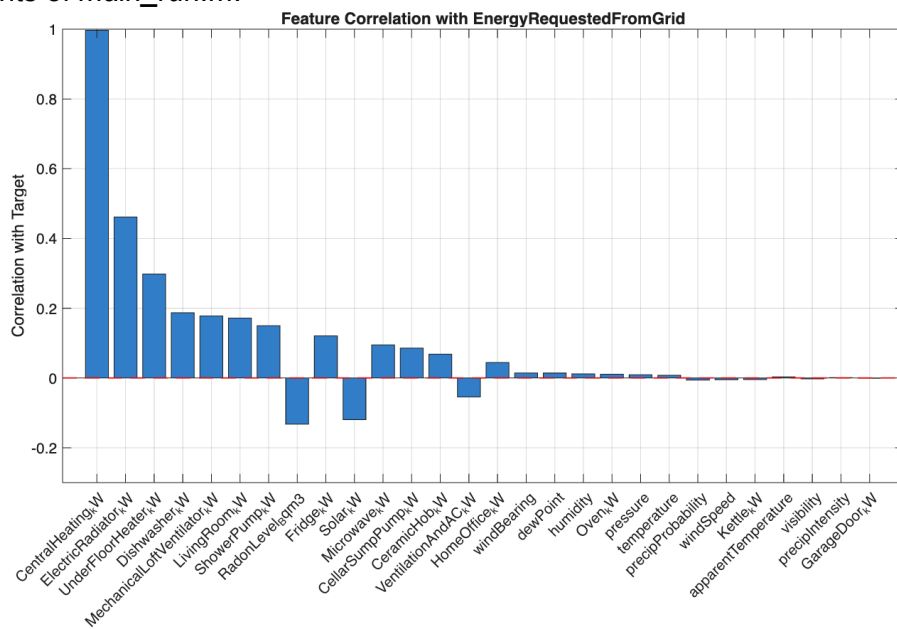


Figure 1 — The Pearson Correlation Coefficients of 28 features

2.4 Data Splitting and Standardisation

The key to machine learning lies not only in training, but also in a method to verify the model, such as training. The dataset was split 80/20 for the train and test set using random hold-out sampling. This method provides an unbiased estimate of the generalisation performance.

After data splitting, a Z-score method (zero mean and unit variance) was again chosen to standardise the train set to avoid data leakage; this process can ensure that the evaluation results fairly simulate unseen data.

3 — Linear Regression (LR) Model Design (Task 2)

3.1 Model implementation

LR is a grey-box model used to predict a target value by calculating the weighted sum of the input features: $\mathbf{y} = \mathbf{a}_0 + \sum \mathbf{a}_i \mathbf{x}_i$, where using least-squares method to minimise errors. Compared to Gradient Descent, LR uses the Normal Equation to achieve the optimal weights analytically in

one single step: $\theta_{\text{hat}} = (\Psi^T \Psi)^{-1} \Psi^T y$ instead of iterating step-by-step toward a minimum. Where Ψ and Ψ^T are the design matrix and the transpose of the design matrix, y is the observed output value in a vector form, and finally θ_{hat} is the estimated weight vector. LR does not require any hyperparameter tuning with a large dataset (~50K samples), but still preserves high accuracy.

3.2 Collinearity

However, if two or more features are highly correlated, a collinearity problem arises, causing the Normal equation ($\Psi^T \Psi$) to become singular, and resulting in small changes in the data can potentially lead to huge and unpredictable fluctuations in the coefficients. To avoid this, feature removal can eliminate it. In Task 1, some features have been removed (e.g., dewPoint, apparentTemp) before the model training. Another approach is applying PCA; however, with 19 features remaining after preprocessing, dimension reduction is unnecessary in this case. Moreover, only three feature pairs exhibit mild collinearity ($|r| = 0.81\text{--}0.91$) and all of them reflect genuine physical meaning, showing the model achieves $R^2 > 0.99$ (See section 3.3), confirming that no further adverse effect is applied on predictive performance.

3.3 Model Evaluation

In section 2.1 the Z-score was chosen to detect outliers. To successfully select the threshold, an iterative method was applied to the MATLAB code ($Z = 3\text{--}15$). The output shows that any thresholds lies on or below $Z = 10$ removed high-demand events aggressively ($R^2_{\text{test}} < 0.98$, $\text{RMSE} > 0.2$), while when $Z = 12\text{--}15$ the results are not only completely stable from R^2 reading, but also minimise the train/test gap. Thereby, $Z = 12$ is finally selected.

The model achieved RMSE (0.0854kW and 0.0860kW) and R^2 (0.9937 and 0.9927) for the train and test sets, respectively. With a negligible gap confirming overfitting problem does not exist. Figures 2.1 & 2.2 below show the predictions vs actual values with a strong linear relationship, and the residuals are concentrated near zero and are symmetrically distributed on the central x-axis.

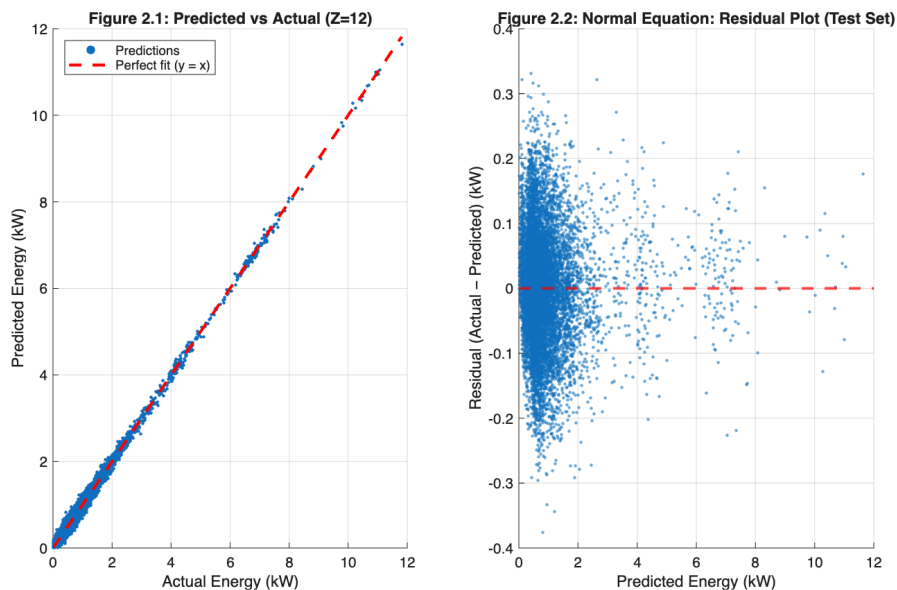


Figure 2.1 & 2.2 - Prediction vs Actual Plot and Residual Plot

4 — Second Model Design (Task 3)

4.1 Model Selection

A more advanced training method should be experimented with by using a more complex and accurate model to better predict the dataset. Hence, a nonlinear regression model should be constructed on 21 features and a 50k+ dataset. The Gradient Boosting (GB) method has been selected; this decision was inspired by the decision tree — it trains trees sequentially on the full data set, then uses residuals of the prior ensemble to iteratively correct all prediction errors until bias has reached minimum instead of variance, such as Random Forest (Bagging method). This offers better controllability at overfitting risk and higher accuracy (unlike a single decision tree). However, GB has a relatively higher computing cost compared to the Bagging method, but still lower than ANN, which also requires significantly more tuning to achieve reliability at this scale of dataset size and type (ANN is more suitable for complex pattern recognition).

4.2 Implementation and Mitigating Overfitting

The same trained and tested set was reused from Task 2 to compare fairly with the new model design. By implementing `fitrensemble` function in MATLAB, two main hyperparameters need to be tuned, including the number of cycles and learning rate. To objectively select the most suitable hyperparameters, a grid search was introduced and performed over 25 combinations. The learning rate controls the total contribution of each tree to the final ensemble — a lower rate offers a better generalisation. Learning cycles control the total correction that the model will make; however, if the correction exceeds the convergence range will result in overfitting. Moreover, a third hyperparameter (MaxNumSplits) was applied but kept at the default setting; it is used to limit individual tree variance.

After training, the optimal configuration of 200 cycles and 0.1 learning rate was finally selected, because it contains the minimum test RMSE and a relatively small train/test gap.

4.3 Nonlinear Regression Results and Comparison

The table below summarises the test and train results for both models:

Model	RMSE_Train	RMSE_Test	R ² _Train	R ² _Test	RMSE_Gap
Linear	0.0854	0.0860	0.9937	0.9927	0.0006
GB	0.0784	0.0839	0.9947	0.9931	0.0055

From the outputs, GB achieved a slightly higher R² and lower RMSE compared with the LR model (Figures 3.1 and 3.2). However, a higher RMSE train/set gap indicates GB reflects the inherent characteristic of sequence tree ensembles to fit the training data more closely than linear models. Despite this, a low train/set gap percentage (GB: 6.6%, Linear: 0.7%) is still considered acceptable, confirming neither model has encountered overfitting problems. Although these results demonstrate that both models can be accurately applied to predicting the grid energy consumption, since the features and target relationship are predominantly linear, GB does not offer a significant performance improvement. Furthermore, GB algorithms are more complex and required to install a specialist MATLAB ToolBox, resulting in a higher computing cost and a lower deployability than a LR model; therefore, LR is finally recommended.

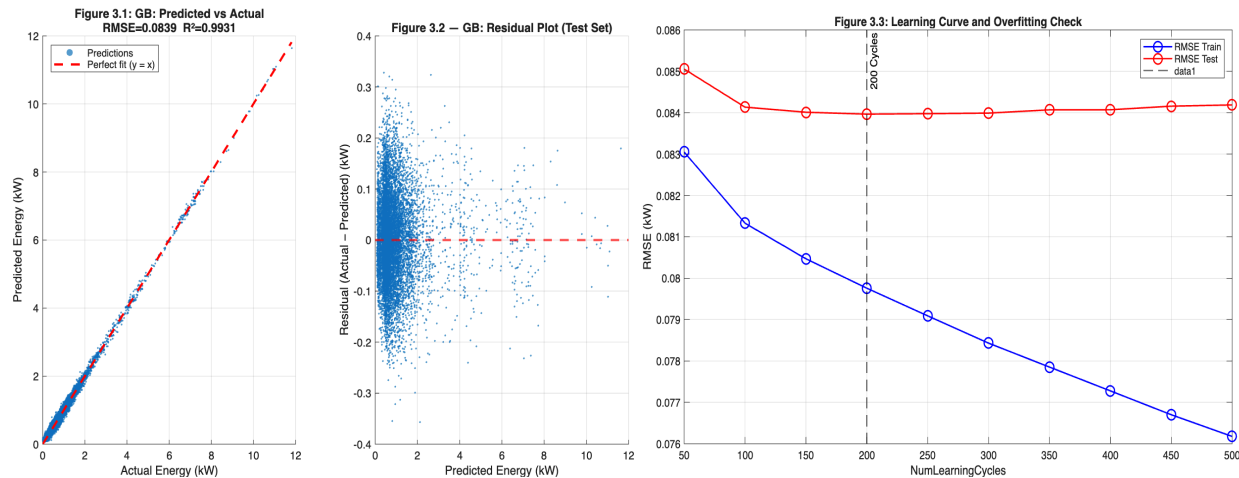


Figure 3.1, 3.2, & 3.3 - Prediction vs Actual Plot and Residual Plot, Learning Curve

5 — Further Discussion

Ethical Issues: When collecting data to train the machine learning models using highly integrated smart devices, the privacy of the data is often highly controversial. McKenna (2012) pointed out that detailed electricity consumption data can reveal living patterns, daily routines, and household behaviors without explicit consent. During model training, no personal information related to customer data should be fed into the model to maximise user privacy and data security. Moreover, relevant laws also list many measures to avoid security risks caused by privacy leaks; they must protect not only user privacy (e.g., **GDPR** [2]) but also that of the power grid company to prevent information leaks caused by malicious attacks (**NIS (2018)** [3]).

Future Challenges: Several limitations and challenges warrant further investigation. First, household behaviour varies seasonally and annually across different regions around the UK, which might require few machine learning models to be trained and retrained periodically to obtain high accuracy predictions. Second, the **Warm Homes Plan (2026)** [4] was initiated recently in the UK, it is accelerating household solar energy adoption across the country. This policy needs to be given significant consideration in the future; however, the electricity storage has not been discussed and taken into account in these machine learning models, as storage will provide further critical features in the training and tuning processes. Finally, the scaling from single-household to national-level forecasts presents aggregation challenges, as individual forecast errors may accumulate rather than cancel each other out at the grid scale.

Reference

- [1]J. Kang and D. M. Reiner, "What is the effect of weather on household electricity consumption? Empirical evidence from Ireland," EPRG Working Paper 2112, Cambridge Working Paper in Economics 2141, University of Cambridge, Cambridge, UK, May 2021.
- [2][UK Government, "Data Protection Act," Gov.uk, 2025.](#)
- [3][UK Government, "The NIS Regulations 2018," GOV.UK, Apr. 20, 2018.](#)
- [4][D. for, "Warm Homes Plan," GOV.UK, Jan. 21, 2026.](#)